$\vec{F} = \langle P, Q \rangle$ (or $\vec{F} = \langle P, Q, R \rangle$)

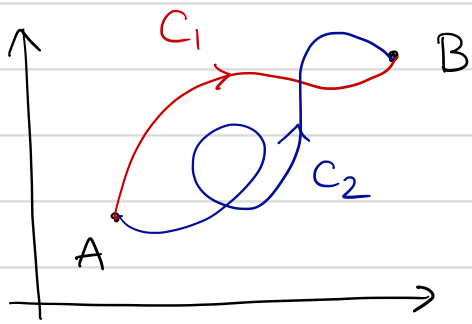
Lecture 14

- FTC for line integrals
- When is $\vec{F} = \nabla f$? (16.3)

Fundamental theorem

$$\boxed{\int_C \nabla f \cdot d\vec{r} = f(B) - f(A)}$$

$\nabla f = \langle f_x, f_y \rangle$ (or $\langle f_x, f_y, f_z \rangle$)



Proof of the theorem:

$$\int_C \nabla f \cdot d\vec{r} =$$

$$= \int_a^b \langle f_x, f_y \rangle \cdot \left\langle \frac{dx}{dt}, \frac{dy}{dt} \right\rangle dt$$

$$= \int_a^b f_x \frac{dx}{dt} + f_y \frac{dy}{dt} \, dt$$

$$= \int_a^b \left[\frac{d}{dt} f(x(t), y(t)) \right] dt \quad \text{Chain rule}$$

$$= f(\underbrace{x(b), y(b)}_{B}) - f(\underbrace{x(a), y(a)}_{A}) \quad \blacksquare$$

Three equivalent properties:

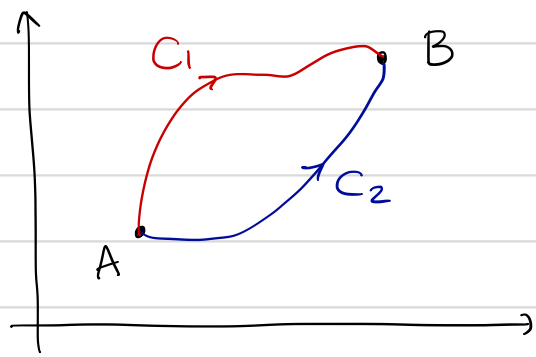
① $\vec{F}$ is path-independent (same work if same end points)

② $\vec{F}$ is conservative $\left(C = \text{closed loop} \Rightarrow \int_C \vec{F} \cdot d\vec{r} = 0 \right)$

③ $\vec{F}$ is a gradient field: $\vec{F} = \nabla f$

Why: (3) $\Rightarrow$ (1), (2) by Fundamental theorem

(1) $\Leftrightarrow$ (2)



$$\int_{C_1} \vec{F} \cdot d\vec{r} - \int_{C_2} \vec{F} \cdot d\vec{r} = \int_{\underbrace{C_1 - C_2}_{\text{closed loop}}} \vec{F} \cdot d\vec{r}$$

(1) $\Rightarrow$ (3): Example soon!

Examples:

1) $\vec{F} = \langle y, x \rangle$ is a gradient field

$f(x,y) = xy \Rightarrow \nabla f = \langle y, x \rangle = \vec{F}$.

2) $\vec{F} = \langle \underbrace{-y}_{P}, \underbrace{x}_{Q} \rangle$ is <u>not</u> a gradient field.

$Q_x = 1 \neq P_y = -1$

Test for gradient fields (2D)

If $\vec{F} = \langle P, Q \rangle = \nabla f$,

then $P = f_x$, $Q = f_y$, so

$$P_y = f_{yx} = f_{xy} = Q_x$$

<u>Claim</u>: If $\vec{F}$ is defined at every point of the plane, and $P_y = Q_x$, <u>then</u> $\vec{F} = \nabla f$ for some f.

<u>Example</u>: Find a so that

$\vec{F} = (4x^2 + axy)\hat{i} + (3y^2 + 4x^2)\hat{j}$

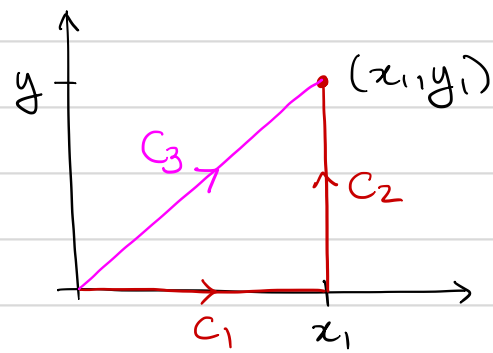
is a gradient field.

$$\left.\begin{aligned} P_y &= \frac{\partial}{\partial y}(4x^2 + axy) = ax \\ Q_x &= \frac{\partial}{\partial x}(3y^2 + 4x^2) = 8x \end{aligned}\right\} \underline{a = 8}$$

Finding a potential:

$\vec{F} = (4x^2 + 8xy)\hat{i} + (3y^2 + 4x^2)\hat{j}$

$= \nabla f$ for some function $f = f(x,y)$.



$$f(x_1, y_1) - \underbrace{f(0,0)}_{\text{constant}} \overset{\text{fund. theorem}}{=} \int_{C_1 + C_2} \vec{F} \cdot d\vec{r}$$

$$= \int_{C_1 + C_2} (4x^2 + 8xy)\,dx + (3y^2 + 4x^2)\,dy$$

[We could also use C_3 instead of $C_1 + C_2$ if that were easier.]

<u>On C_1</u>: $\begin{cases} x = x \\ y = 0 \end{cases} \Rightarrow \begin{aligned} dx &= dx \\ dy &= 0 \end{aligned}$

$$\int_{C_1} \vec{F} \cdot d\vec{r} = \int_0^{x_1} (4x^2 + 0)\,dx = \frac{4}{3}(x_1)^3$$

<u>On C_2</u>: $x = x_1 \quad y = y \quad dx = 0$

$$\int_{C_2} \vec{F} \cdot d\vec{r} = \int_0^{y_1} 3y^2 + 4x_1^2 \, dy$$

$$= y_1^3 + 4x_1^2 y_1$$

<u>Conclusion</u>

$$f(x_1, y_1) = \int_{C_1 + C_2} \vec{F} \cdot d\vec{r} + \underbrace{f(0,0)}_{\text{const}}$$

$$= \frac{4}{3} x_1^3 + 4x_1^2 y_1 + y_1^3 + \text{const.}$$